

Literature Citation Verification System Based on DeepScientist

Project A — Engineering Track: Evidence Chain Module

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The Problem: Trust Deficit in Research Agents

Research Agents Have a Credibility Gap

LLM-based research agents (DeepScientist, OpenClaw) produce **professional-looking reports** with citations — but users cannot answer three critical questions:

- ❶ **Which conclusions are backed by real sources?** Citations are untraceable
- ❷ **Which are model inferences or hallucinations?** No automated detection exists
- ❸ **Which have no evidence at all?** Wording often stronger than evidence justifies

Real-World Consequence

In our baseline testing with unmodified DeepScientist: a single report contained **3 fabricated citations** that passed visual inspection. The agent confidently cited “Smith et al. 2023” for claims — but that paper does not exist in Semantic Scholar’s database of 200M+ papers.

Why This Matters: The Hallucination Problem

The Scale of the Problem

- LLMs generate **3–27% fabricated references** in academic writing tasks (Walters & Wilder, 2023; Bhatt, 2024)
- In scientific peer review, **up to 69% of GPT-4 generated reviews** contain hallucinated citations (Liang et al., 2024)
- These fabrications are **indistinguishable from real citations** without external verification
- For research agents that autonomously generate proposals and ideas, this risk is amplified

Our Response

- **Every claim** extracted from a source paper is verified against a **real API** (not an LLM)
- **3-source fallback**: Semantic Scholar (200M+ papers) → arXiv (2.4M open-access) → Crossref (150M+ records)
- Results are **scored**, not estimated: verdict + confidence + evidence snippet
- Final report includes an **auditable evidence chain** that cross-references every citation

System Architecture: Before & After

Before: Original DeepScientist

QQ User $\rightarrow$ Agent (LLM) $\rightarrow$ Report

No verification
No audit

No evidence IDs
C+ grade

After: Our System

QQ $\rightarrow$ crossdisc_idea (C) $\rightarrow$ parse_pdf (A)



verify_claim $\times N$ (A) $\rightarrow$ score_batch (B)



render_report (B) $\rightarrow$ record+memory (C) $\rightarrow$ QQ

+ 5 new modules + 3 MCP namespaces
+ 139 tests $\geq$ B- grade

A (Tool): claim_extractor, semantic_verifier, HARD STOP RULE **B (Scoring):** traffic_light, render, MCP 5-tool reg **C (Integration):** crossdisc_idea, QQ evidence, memory schema

FactCheck Pipeline: 5 Phases, 6 Mandatory Tools

Phase 0 — PDF Parsing (A) PyPDF2 → regex detection of “REFERENCES” → split by [N] / [Author, Year] → Unicode quote title matching → truncation at 200K chars / 40 claims. Handles IEEE, APA, and named-author formats.

Phase 1 — Verify (A) *323 calls across 6 quests* (1) Semantic Scholar → (2) arXiv Title API → (3) Crossref DOI search. Returns verdict + confidence + evidence snippet.

Phase 2 — Score (B) *7-branch RYG logic*

supported conf ≥ 0.7	● Green (correct)
contradicted conf ≥ 0.7	● Red (wrong)
all other combinations	● Yellow (uncertain)

Phase 3 — Render (B) RYG summary table + per-claim detail cards + Evidence Chain table (E00X ↔ C00X).

Phase 4 — Persist (C) *36/36 calls*
artifact__record → evidence_store.
memory__write → episodes.

Data Flow Claim → VerificationResult → ScoredClaimResult → FactCheckResult

Enforcement — 2 Layers L1: 5-State Machine (SKILL.md) with exit checks. L2: HARD STOP RULE (builder.py).

Evidence Chain: Making Every Claim Auditable

3 Evidence Types (E00X)

Source	ID Format
QQ Text	E001
QQ Image	E001-img
QQ PDF	E001-pdf

All entries: SHA256 hash. PDF: +page count, +text sidcar.

Report Structure §1 FactCheck, §2

Evidence Chain, §3–5 Bridges, §6 Caveats.

Audit: `audit_report()`

- Scans report for evidence ID patterns (E, E-img, E-pdf)
- Cross-checks cited IDs vs. `evidence_store`
- Detects fake IDs, unused evidence, bare claims
- Outputs PASS or WARN verdict

Three-Tier Annotation

[E001] Evidence-supported (●)

Inferred Logically derived

Needs Verif. Insufficient evidence

Live Demo: What Actually Happens

User sends via QQ:

/crossdisc_idea
[Attaches: pFedAFM.pdf, 14 pages]

Bot processes (2–5 min):

- 1 Downloads PDF → extracts text via PyPDF2
- 2 Calls parse_pdf → **40 structured claims**
- 3 Calls verify_claim × **85 times** across S2/arXiv/Crossref
- 4 Calls score_batch → **G17 Y18 R5 (FAIL)**
- 5 Calls render_report → colored Markdown
- 6 Calls artifact__record + memory__write

Bot returns via QQ:

FactCheck Score: **FAIL**

● 17 correct ● 18 uncertain ● 5 wrong (40 claims)

5 Cross-Discipline Bridges:

Medical diagnosis → Autonomous driving
→ Climate modeling → FinTech → NLP

*Full evidence chain table with
E001–E040 ↔ C001–C040 mapping*

Test Results: 6 Quests Across 6 Disciplines

Quest	Discipline	Paper	Claims	Verif.	Score
027	ML/Federated	pFedAFM (14pp)	40	85	G17 Y18 R5 FAIL
028	Medical Edu.	Subinternship Dir.	8	27	G1 Y6 R1 FAIL
029	Chemistry	DFT Metal Clusters	1	7	G1 Y0 R0 PASS
030	Optimization	SGD Convergence	40	105	G10 Y28 R2 FAIL
031	Physics	LIGO PEMcheck	40	94	G2 Y37 R1 FAIL
032	ML Survey	FL Survey	1	5	G0 Y1 R0 WARN

Aggregate Metrics Across All 6 Quests

139 tests all pass | **996 events** logged | **323 verify_claim** calls | **36/36**
mandatory tools completed

6 disciplines: CS · Medicine · Chemistry · Mathematics · Physics · Survey

Agent Compliance Journey: From 0 to 36/36

Iter.	Approach	Result	Lesson
v1	SKILL.md suggests	0/6 tools	LLMs ignore “please”
v2	MUST + code examples	10/40 verify, 0 score	Agent still cuts corners
v3	5-State Machine + exits	Partial fix	Exits help but bypassable
v4	HARD STOP RULE	36/36	Framework > Prompt

Key Insight: Prompt-Level vs. Framework-Level Enforcement

SKILL.md is advisory — the agent can still choose to hand-write scores. builder.py's HARD STOP RULE is mandatory — injected into system context and cannot be ignored. **Key lesson:** agent design must distinguish advisory from mandatory constraints.

Failure Cases: What Breaks and How We Handle It

- ❶ **Verifier False Positives (LIGO Q031)** C036 (GW190707 event) flagged ● contradicted. Semantic Scholar matched a different paper with overlapping author names. **Handling:** Report explicitly states “almost certainly a false negative — verifier matched wrong paper.”
- ❷ **Abstract-Only Verification Limits (SGD Q030)** C018/C031 (mathematical convergence proofs) unverifiable from abstracts alone. **Handling:** Claims marked ● uncertain; caveats note “full-text access required for theorem verification.”
- ❸ **PDF Parser Cross-Discipline Failure (Medical Q028)** Medical papers indexed via PubMed, not arXiv — `parse_pdf` returned 0 structured claims despite successful text extraction. **Handling:** Agent auto-fell back to `bash_exec` manual extraction; evidence table marked `extraction_method: bash_fallback`.
- ❹ **DeepSeek API Version Incompatibility (early, fixed)** Claude Code $\geq 2.1.154$ sends system role messages unsupported by DeepSeek — 80% of runs crashed. **Fix:** locked Claude Code to v2.1.153, added `--strict-mcp-config`, filtered system Python from MCP PATH.

Boundary Cases: Proving Robustness at the Edges

Case	Quest	What We Proved
Minimal (1 claim)	Q029	System correctly outputs PASS; no crash on sparse input
Max uncertainty	Q031	37/40 🟡; honestly explains abstract-only limitation
Non-standard refs	Q031	LIGO astro-ph format; system marks as not_found, not hallucinated
Internal proofs	Q027/030	Theorems have no external citation target; correctly flagged
Multi-citation	Q027	[20]--[24] dash-range parsed into 5 individual markers
Fake ID	test	audit_report correctly identifies E999-img as non-existent

Coverage: Normal case (Q027, Q030) + Degraded case (Q028) + Edge case (Q029) + Negative case (fake IDs) + Cross-domain case (Q031)

Key Technical Decisions

Decision	Choice	Rationale
Base system	DeepScientist	QQ connector + MCP framework + skill loader
PDF parser	PyPDF2	Handles 90% of academic PDFs, already installed
Verifier API	Semantic Scholar	Free, 200M+ papers, no API key needed
Fallback chain	S2 → arXiv → Crossref	3 complementary indexes, different metadata
Scoring logic	7-branch RYG	All verdict × confidence combinations covered
Enforcement	Prompt + Framework	SKILL.md suggestive, builder.py mandatory

What We Did NOT Build (and Why)

We deliberately scoped out full-text semantic entailment (requires paid API access) and LLM-based claim extraction (would introduce circular dependency — verifying LLM output with another LLM).

Before / After: What Changed

Dimension	Before	After
Message Recording	QQ logs only	E00X store (SHA256 + metadata)
Citation Annotation	None	[E001] + RYG 3-color labels
Reference Verification	LLM-only (hallucination risk)	S2 + arXiv + Crossref API
PDF Evidence	Not supported	Pages + SHA256 + text sidecar
Audit Capability	None	audit_report fake-ID detection
Experiment Logging	None	memory_write episodes schema
Test Coverage	0 tests	139 tests / 100% pass
Cross-Discipline	Single domain	6 disciplines validated
Tool Chain	C+ (prompt only)	≥ B- (independent tools)

Current Limitations

- Abstract-only: most ● would be resolvable with full-text access
- Verifier false positives: $\sim 5\text{--}10\%$ ● from title-matching errors
- PyPDF2 weak on scanned/image-based PDFs (requires OCR preprocessing)
- Network-dependent: offline mode not supported
- No video/audio evidence support

Future Improvements

- Full-text verification via arXiv open-access PDF parsing
- Local model fallback: Ollama + sentence-transformers for offline mode
- Structured NLP pipeline: replace regex with transformer-based claim extraction
- Multi-connector: extend to WeChat, Discord, Telegram
- Result caching: avoid duplicate API calls for same (claim, paper) pairs

Team Contributions

Member & Role	Scope	Key Deliverables
Miaoyi Liao	4 new + 3 mod files	claim_extractor, semantic_verifier (3-API chain), HARD STOP RULE,
Person A: Tool Layer	~1,500 loc	evidence-chain store + runner instrumentation
Xiaoyu Xiong	3 new + 2 mod files	traffic_light (7-branch, 24 tests),
Person B: Scoring	~1,200 loc	factcheck_render, MCP 5-tool registration, 6 Quest execution + all debugging
Yuxiang Liu	3 new + 8 mod files	crossdisc_idea SKILL.md (5-Phase SM + state machine), QQ evidence
Person C: Integration	~800 loc	recording, memory schema, daemon+runner fixes, 16 tests

Cross-Collaboration: The Phase 5 Enforcement Story

A provided HARD STOP RULE (builder.py) — **mandatory**. B rewrote Phase 5 as 5-State Machine (SKILL.md) — **structural**. C defined memory/artifact schema + evidence chain template — **auditable**. Three layers, one goal: agent cannot bypass the toolchain.

Summary: From Promise to Proof

We built a system that verifies what LLMs claim.

Scale	Rigor	Impact
6 disciplines	139 tests / 100% pass	C+ $\rightarrow$ $\geq$ B-
6 quests completed	323 API verifications	36/36 mandatory calls
996 events logged	3-source fallback chain	4 failure cases analyzed

The system transforms research agents from **unverifiable storytellers** into **auditable evidence processors**. Every claim has an ID. Every ID has a source. Every source is checked.

Code: https://github.com/leoplasture/STA304_Final_Project **Report:**
docs/Final_Project_Report.md